

Week 2: From Strategy to Implementation

Transform the Researcher into a full planner and code generator, setting up a Test-Driven Development (TDD) foundation for the Orchestrator.

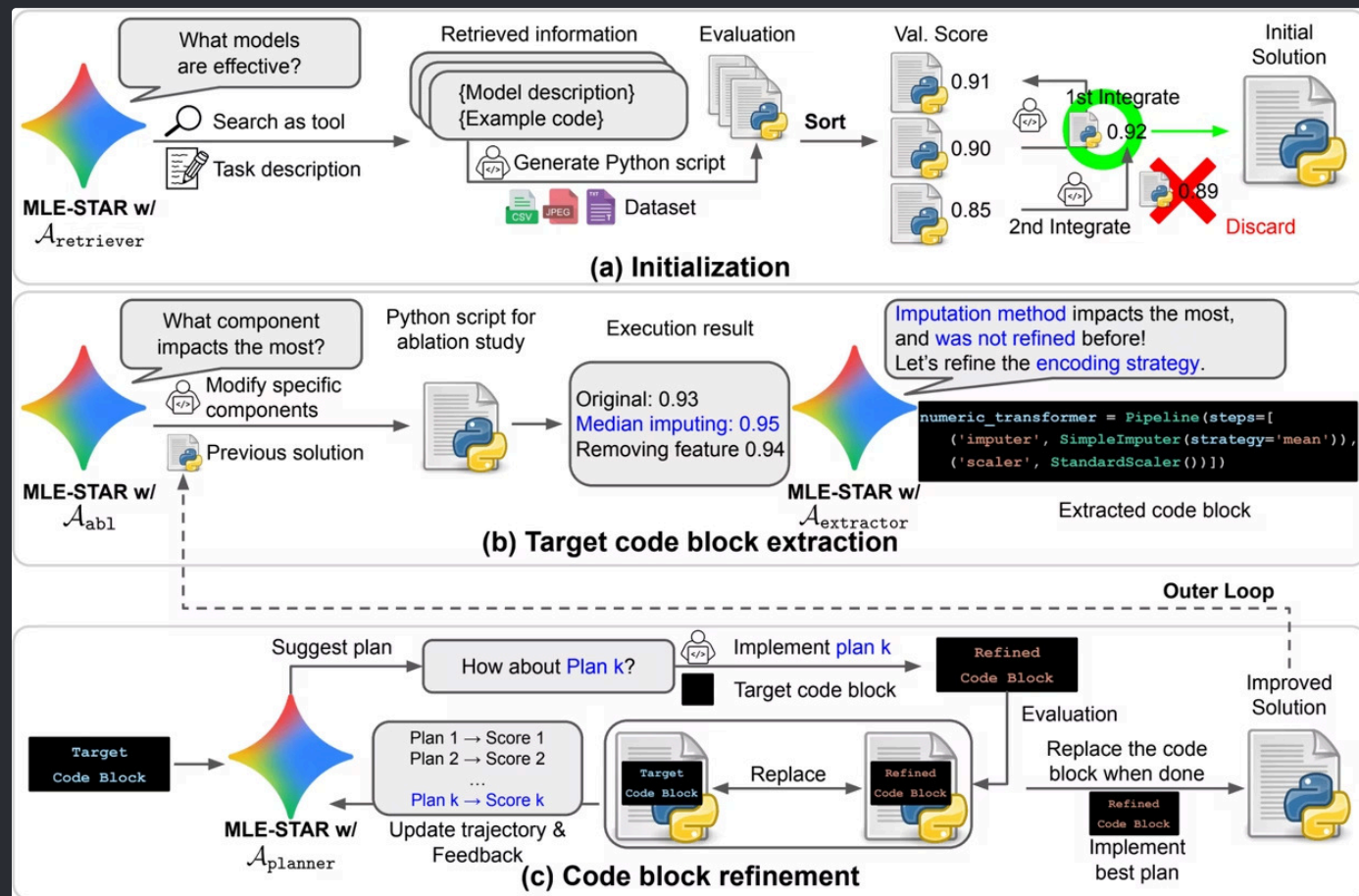
Key Concept: Structured output parsing and Test-Driven AI Development.

 by Alwin Lin

Welcome to week 2

- Review from week 1
- The Manager's Dilemma
- Test-Driven Development (TDD)?
- Engineering the Prompt: Forcing Structure
- Q&A and Homework Assignment

From Week 1



System Prompt, Skills, Tools:

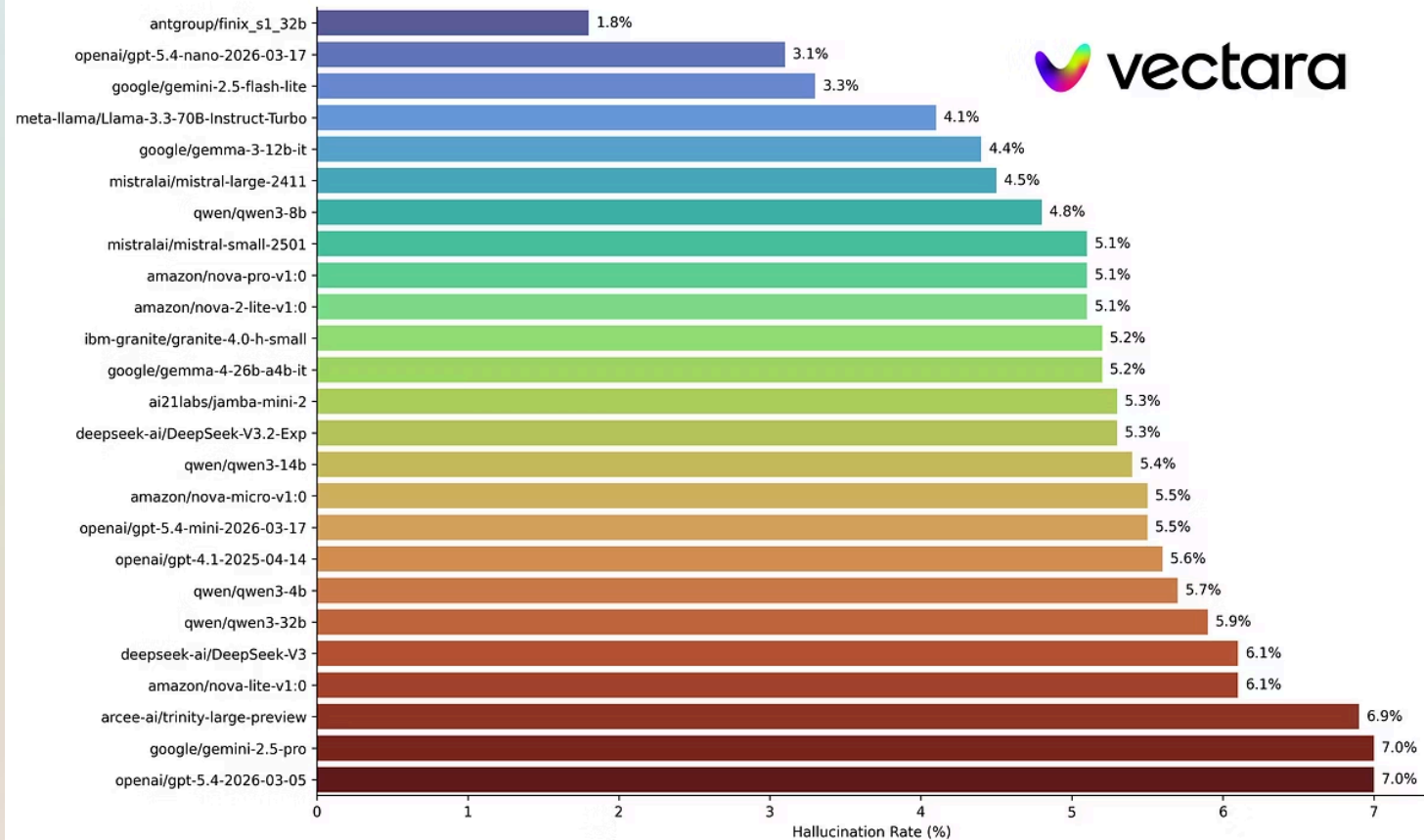
- What are you ?
- What can you do?
- How do you do that?

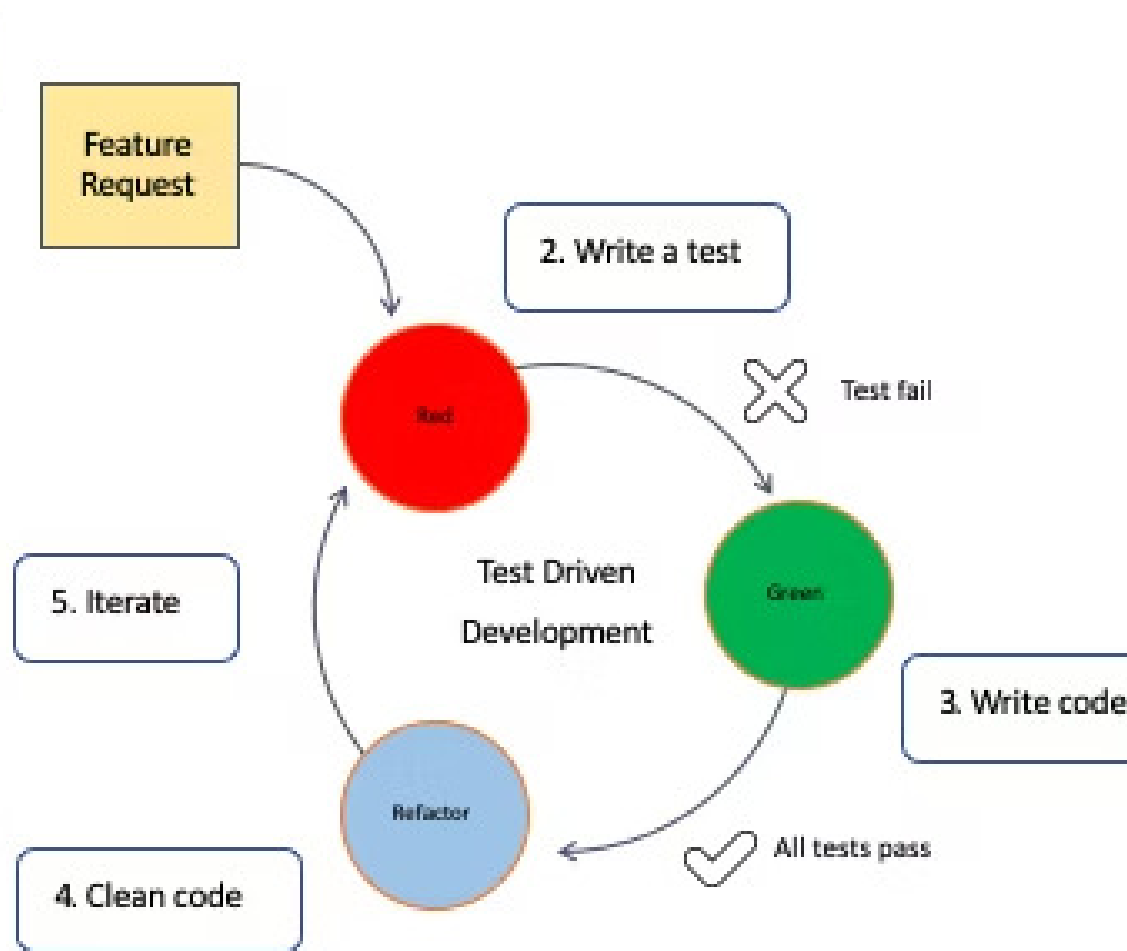


The Manager's Dilemma

- LLMs Hallucinates
- LLMs can't/refuse tell what is wrong in code
- Standardizations

Grounded Hallucination Rates for Top 25 LLMs





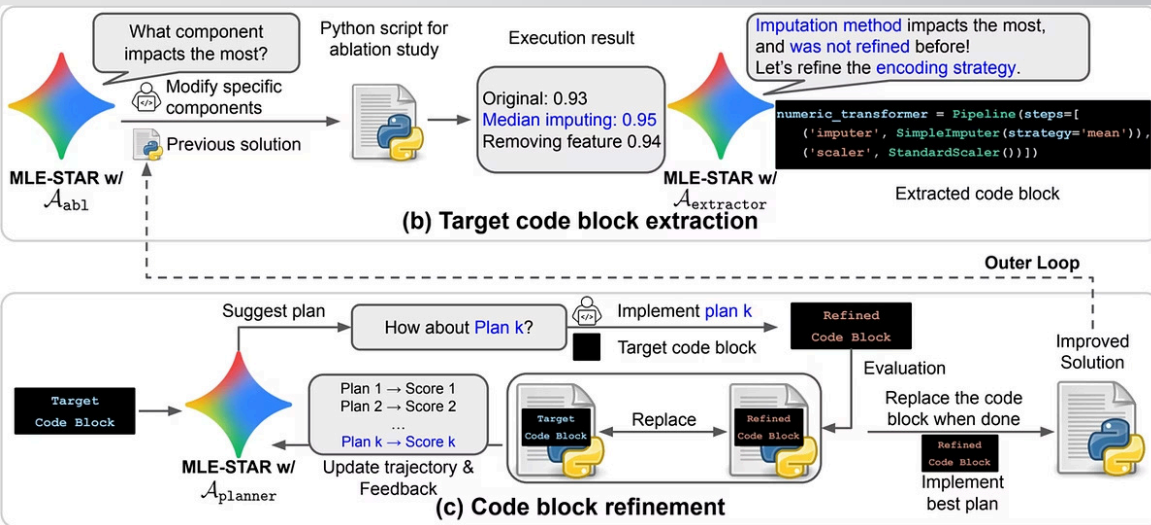
Why Test-Driven Development (TDD)?

- Feature Request
- Test
- Code
- Clean up

MLE-star Framework (Machine Learning Intelligence)

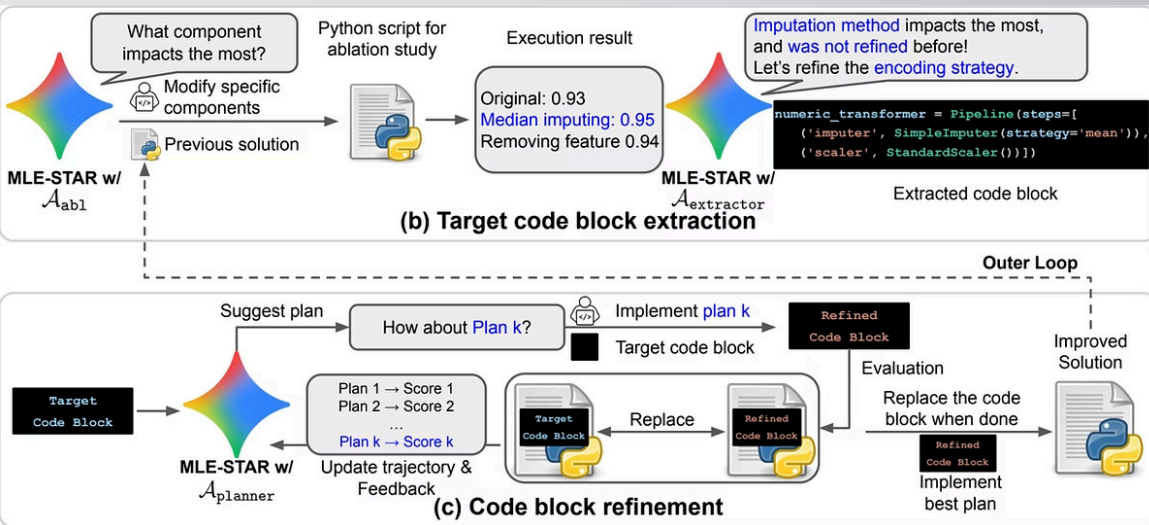
Eliminating Silent Failures in Machine Learning Pipelines

- Data Leakage Checker
- Data Usage Checker
- Search-Based Validation
- Automated Debugging & Traceback Analysis



MLE-star Framework (Machine Learning Intelligence)

- Higher ML confidence score
- Target Change code blocks
- Block Change and Benchmark
- Repeat



Engineering the Prompt: Forcing Structure

- LLMs are mostly chatty
- Reinforcements of structural outputs
- Options: Skills, System Prompts, Hardcode

Normal Claude (69 tokens)

"The reason your React component is re-rendering is likely because you're creating a new object reference on each render cycle. When you pass an inline object as a prop, React's shallow comparison sees it as a different object every time, which triggers a re-render. I'd recommend using `useMemo` to memoize the object."

Normal Claude

"Sure! I'd be happy to help you with that. The issue you're experiencing is most likely caused by your authentication middleware not properly validating the token expiry. Let me take a look and suggest a fix."

Caveman Claude (19 tokens)

"New object ref each render. Inline object prop = new ref = re-render. Wrap in `useMemo`."

Caveman Claude

"Bug in auth middleware. Token expiry check use `<` not `<=`. Fix:"

Looking Into Week 3:

- Setting up the environment
- Auto Pipe runs
- Benchmarks